Empirical Software Engineering

Write your answers directly on these pages (there's always a risk that loose papers disappear)—use the back also if possible. I'll be at the written exam at 10.00 for questions.

On February 2 at 10.00, you are welcome to Richard's office (Room 4123, 4th floor in the EDIT building at Campus Johanneberg) to complain about the grading.

Before the meeting you *must* send Richard an email clearly pointing out where you think the error is, what you wrote, and why you believe the grading was not correct. If I don't receive such an email before 10.00 on February 2, then I will not meet with you.

Experience is simply the name we give our mistakes

— Oscar Wilde

Grade 3: 40 points; ~50%
Grade 4: 56 points; ~70%
Grade 5: 72 points; ~90%
Maximum: 80 points

Question 1 :

(**8p**) In *your* opinion, what key steps are compulsory when conducting Bayesian data analysis? Please explain each step one takes when designing models so that we can place *some* confidence in the results.

You can either draw a flowchart and explain each step, or write a numbered list explaining each step.

(It's OK to write on the backside, if they haven't printed on the backside again...)

Question 2 :

(**6p**) Write down two complete model designs below. One model should maximum overfit, and one model should maximum underfit. Be careful using math notation.

Question 3 :

(**3p**) Please provide **an example** where you **argue** epistemological and ontological reasons for selecting the Poisson distribution.

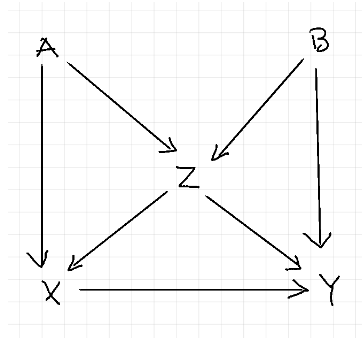

Figure 1: A messy Directed Acyclic Graph.

Question 4 :

(**6p**) Look at the DAG above. We want to estimate the **direct** effect of X on Y. Which variable(s) should we condition on **and why**?

Question 5 :

(**8p**) What is the **purpose** and **limitations** of using computer simulations and experimental simulations as a research strategy?

Question 6 :

(**4p**) Below you see a Generalized Linear Model.

$$
\begin{aligned}
y_i &\sim \text{Poisson}(\lambda) \\
\text{f}(p_i) &= \alpha + \beta x_i
\end{aligned}
$$

What should f() be in this case, and what is its inverse? (2p)

Does f() somehow affect your priors? If yes, provide an example. (2p)

Question 7 :

(**4p**) Explain the four main benefits of using multilevel models.

Question 8 :

(**4p**) When diagnosing Markov chains, we often look at several diagnostics to form an opinion of how well things have gone. One common diagnostics is $\widehat{R}$.

What is a commonly accepted threshold value for $\widehat{R}$ (1p)?

Can the following hold: $\widehat{R} < 1.0$? If yes, why (1p)?

How is $\widehat{R}$ measured (2p)?

Question 9 :

(**10p**) We often differ between reliability and validity concerning surveys,

1. What is the difference between the reliability and validity in survey design? (2p)
2. Name and **describe** at least two types of reliability in survey design. (4p)
3. Name and **describe** at least two types of validity in survey design. (4p)

Question 10 :

(**8p**) When we analyze open questions in survey designs we can't easily use Bayesian data analysis due to the qualitative nature of the data.

List and explain the four common steps when analyzing open questions, according to what we have covered in the course.

Question 11 :

(**5p**) Below, please write out the design of a **complete** Bayesian model that includes hyperparameters and hyperpriors, i.e., a multilevel model. Please use realistic priors according to your knowledge and skills, and be careful with the math notation.

Question 12 :

(**4p**) You have designed a model where you have a general intercept and then a varying intercept with two levels. The two levels indicate two design levels in a software system: 'detailed' and 'coarse'. After having sampled the model using Hamiltonian Monte Carlo you want to see how often 'detailed' has a higher average than 'coarse.'

Explain which steps you would take to show this? (2p)

How would you argue that the difference between 'detailed' and 'coarse' is not there by chance? (2p)

Question 13 :

(**10p**) Explain the following ten concepts:

1. Multicollinearity (1p)
2. Omitted variable bias (1p)
3. Post-treatment bias (1p)
4. Collider bias (1p)
5. Confounding (1p)
6. Regularizing prior (1p)
7. Maximum entropy (1p)
8. Absolute and relative diffs (1p)
9. Over-dispersion (1p)
10. Zero-augmentation (1p)